TREETABLE

What if nature was the cloud?

Today's internet lives in buildings full of machines that wear out. The living cloud is forests, reefs and soil — listening, thinking, remembering — and every living system keeps the credit.

Treetable LLC · T.R. Price · terexmaps@gmail.com · terexmaps.com

THE IDEA

Nature stops being the data source. It becomes the provider.

Sense

Forests, reefs and soil already emit signals — birdsong, plant clicks, water chemistry. Small boxes listen without touching anything.

Think

Answers run through maps of real animal nerves — and, piece by piece, through living systems themselves.

Remember

Every answer ships a receipt anyone can re-check. Later, storage moves into nature's own media.

Not a data center built next to nature — the cloud *is* nature. Machines are just scaffolding, and each generation needs less of them.

RUNNING NOW

What already works LIVE NOW

Receipted compute API

Deterministic tasks over 15 EM-derived connectome wirings. Every call returns a replayable receipt — verified bit-identical across two implementations (930/930 checks).

`connectome-compute-api.terexmaps.workers.dev`

Federated sensing

26,000+ allied devices already listening — BirdWeather detections stream through our live endpoint right now. Their devices, their counts, labeled as theirs.

`GET /federation`

Visitor nodes

Any browser becomes a listener: the permission prompt is the consent, analysis stays on-device, only receipts travel.

Open evidence layer

AGPL stack — quorum witnessing, mutual-credit settlement, guardian custody. 121/121 e2e checks under Guile.

HEADING OUTDOORS

Built — field test next BUILT

EcoCell One · ~$35–45

Universal listener — every head's electronics on every device: camera, air mic, env, hydrophone front-end, ultrasonic MEMS. Battery by default, solar optional. Manufacturable KiCad, DRC-clean, pick-and-place files ready.

Reads *and* writes

The node is bidirectional: it reads living systems and — only on joint guardian+operator consent — writes back, gently. Nothing attaches, nothing enters soil. Every emission logged and receipted.

Backhaul is federated too: LoRa mesh, LoRaWAN, APRS, Meshtastic — thin seams over existing radios, ~5% custom code total.

THE CONSENT CONTRACT

Four rules, enforced in code

1 · We never attach

No sensors on plants or animals, nothing in soil. Listening from a distance is the default.

2 · Nothing is hidden

Every reading and answer gets a receipt anyone can replay.

3 · Nature has a voice

Each ecosystem has a human guardian with custody. Refusal is a first-class value — the API says "no" out loud.

4 · Two yeses to write

Writes need guardian + owner consent, stay minimally disruptive, and refused lanes (hop / compute / engineered) refuse outright.

Not poster promises — the lane gate (`lane-gate.rego · offers.scm`) refuses what breaks them.

WHY NATURE

Green improves. Grey decays.

Machines wear out

Every grey solution degrades — silicon ages, buildings depreciate, batteries die.

Living systems heal

Forests self-repair, self-power, and get better at surviving every year — for free.

PROVEN BY SCIENCE Sparse biological wiring beats dense compute at scale — **measured**: medulla (40k neurons) 1.17e-1 J vs dense fp32 8.30 J (~70×). The degree-matched shuffle control refuted our own flattering interpretation — and we published it.

Ecosystems are already worth more alive — air, water, soil. If they also join humanity's knowledge work, the alignment deepens.

THE PATH

Step by step, work moves to nature

LIVE NOW **Receipts + compute over real connectomes** — replay-verified API, federation streaming, visitor nodes.

BUILT **Field listeners** — EcoCell One fleets witness each other; first living quorum.

NEXT FRONTIER **Nature does more of the job** — organism-bound memory, living compute — each step consent-gated, proven by receipts before it ships.

LOCKED UNTIL PROVEN **What we refuse** — relaying strangers' traffic, unwitnessed claims, any write without two yeses.

Each generation must prove with receipts that it uses less power, space and time than the last.

THE MARKET MARKET RESEARCH

Buyers pay for data they can't verify. We sell proof — then the cloud itself.

$380B+

cloud + environmental data markets combined — cited research, not our measurement

$5–40B

biodiversity credits & MRV — where integrity demand is rising fastest

$35–45

our listener BOM vs $600+ incumbents — hardware becomes distribution

Revenue ladder: verification audits → replay-verified API tiers → hardware via crowdfunding → receipted MRV → mutual-credit settlement margin.

TRACTION

Already shipped

- **Compute API live** — replay-verified tasks over 15 connectomes; TS port bit-identical to the Python engine (930/930)
- **Evidence layer open** — sensing → quorum → receipt → settlement, 121/121 e2e, AGPL
- **Federation live** — 24 adapter seams; BirdWeather detections served in production
- **EcoCell One** — fab-ready Gerbers, BOM, pick-and-place; firmware passes e2e
- **NSF SBIR/STTR pitch submitted** (#00131003) · Crowd Supply application filed
- **Benchmark published** — first measured efficiency table over real connectomes, with honesty tiers and an unfavorable control included

JOIN IN

The cloud that plants would join willingly.

Host a box

A ~$40 listener near your garden or land. Waitlist open.

Build with us

AGPL stack, live API, docs — github.com/treetable/living-resources

Fund the frontier

Verification revenue funds the field pilot; hardware follows.

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